Solutions to Exploration 4.1

1. Obs units: Games (41 of them), Variables: Sell-out crowd (y/n) binary categorical, Won game? (y/n) binary categorical
2. Higher win % in front of smaller crowd. Small crowd: 12/23 = 0.522 Sellout crowd: 3/18 = 0.167
3. Yes, the proportions are fairly different from each other.
4. Explanatory = sell out and response = won game,
5. Perhaps, sell-outs happen more often when playing good teams. If so, this would mean that the Thunder are more likely to lose. Thus quality of opponent is a possible confounding variable.
6. Quality of opponent (High/low), if quality of opponent is high, sell out is more likely, if quality of opponent is high, winning the game is less likely.
7. High quality: 13/22 = 0.591, Low quality: 5/19 = 0.263. More likely to sell out against a strong (high quality) opponent
8. High quality: 4/22 = 0.182, Low quality: 11/19 = 0.579. More likely to win against a weak (low quality) opponent
9. The analysis confirms the relationship suggested in numbers 5 and 6, quality of opponent is a confounding variable, it is associated with both the sellout crowd and whether the Thunder won the game.
10. Having a sell-out did not CAUSE the Thunder to play worse. Another variable, quality of opponent, was associated with both sell-outs and winning percentage. Higher quality opponents led to sell-outs AND the Thunder losing games. So, yes, the Thunder did lose more sell-out games, but this was likely due to the quality of opponent.

Solutions to Exploration 4.2

1. No, because then sex of the subject (male/female) would be confounded with the treatment strategy (elevating and lowering), and if we saw a difference in groups we wouldn't know which explanation is more plausible.
2. Proportion Male in both groups is 67% (8/12)
3. No, there are equal proportions of males in each group as there are equal proportions of females in each group. The difference could be due to some other variable that wasn't controlled for.
4. Randomly assign the subjects to the two different groups.
5. Expect 67% (2/3) to be male in the elevating strategy and 67% (2/3) to be male in the lowering strategy. It won't always be this way, just on average.
6. Answers will vary. One example: prop men in Group 1 = 75% (9/12), prop men in Group 2 = 58.3% (7/12), difference is 16.7%
7. Answers will vary. Sometimes it will be the same, but most of the time it will not.
8. 0
9. No, it won't always be equally distributed. However, there is a tendency to have the same proportion of men in both groups.
10. Yes, on average it is the same; the average heights of the two groups tend to be similar when randomly assigning people to two groups.
11. Yes, on average it is the same; the proportion of people with the gene tends to be similar when randomly assigning people to two groups.
12. Yes, on average it is the same, the distribution of the *x*-variable seems to be similar between the two groups.
13. Cause-effect, because random assignment has eliminated other possible explanations. One would need to know more about the sample of people (who they represent, etc.) in order to draw the conclusion to a broader population.
14. Since random assignment was used all confounding variables should be neutralized, or equalized, between the two groups, so any significant difference can be attributed to the treatment (heightening or lowering strategy), suggesting a potential cause-and-effect relationship.
15. Obs units: Students, Exp variable: Whether wrote in cursive (Y/N)-categorical, Resp variable: score on SAT essay (quantitative)
16. Observational study = the researchers did not intervene and assign writing styles (cursive or print) to the subjects.

17. No, can't conclude that writing style caused performance. It could be that kids with better ability levels are more likely to write in cursive. Potential confounding variable: ability of student.
18. This study deals with the potential confounding variable of student ability level, by ensuring that both groups have students with the same ability levels,
19. Potentially, because we've dealt with potential confounding variables.
20. If they did, it might impact the scores that they gave.
21. The people measuring values of the response variable should be blind to group assignment so that they don't let their knowledge impact their measurement.